

Literature-Implied Status of Shearlet Riesz-Basis Existence

Source: arXiv:1404.1690

FA/Banach run `fa_banach_001`, lane 7

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Status

`literature_implied_answer`, scoped as: broad existence positive; compact cone-adapted subclass open. This packet records a theorem-to-question identification, not a new proof.

Source question

Ma and Petersen study admissible compactly supported separable cone-adapted shearlet systems in [1]. In Section 5, “Beyond linear independence,” PDF page 18, immediately after their Lemma 5.2, they write:

The question about the existence of ω -independent shearlet frames, or equivalently shearlet Riesz bases is a delicate question for future work.

Their Lemma 5.2 states, in particular, that a frame is a Riesz basis if and only if it is ω -independent.

Supporting theorem and identification

Führ and Maus construct what their abstract calls the first known shearlet Riesz basis [2]. Their Corollary 4.3, on PDF page 9, states that for any rational $a > 1$ and $b > 0$ there exist $\psi \in L^2(\mathbb{R}^2)$ and $\Lambda \subset \mathbb{R}^2$ such that

$$\left(D_{A_{a^k}} D_{S_{mb}} T_{\lambda} \psi \right)_{k, m \in \mathbb{Z}, \lambda \in \Lambda}$$

is a Riesz basis of $L^2(\mathbb{R}^2)$. Hence it is also an ω -independent shearlet frame. This gives a positive answer to the literal broad existence question in [1].

The supporting paper does not cite Ma–Petersen or state that it is answering their question. The relation is therefore an agent-identified implication, not an explicit same-problem claim by the supporting authors.

Scope limitation

The supporting construction is outside the exact class studied in the source paper. It uses a wavelet-set generator, with $\hat{\psi}$ proportional to the characteristic function of a bounded frequency set, and the translation set Λ is a quasicrystal. It therefore does not construct a spatially compactly supported, separable, cone-adapted shearlet Riesz basis with the source paper’s regular lattice sampling. Indeed, Führ–Maus Remark 4.5, also on PDF page 9, only says that cone-adapted shearlet

Riesz bases are expected. The contextual compactly supported cone-adapted/lattice subproblem remains open on the evidence reviewed here.

Search evidence and review recommendation

The run indexes were searched using both arXiv ids and shorter keyword combinations involving shearlets, Riesz bases, omega independence, cones, and compact support. Local primary-source comparison and bounded web/arXiv searches found the decisive Führ–Maus construction but no later theorem closing the compactly supported cone-adapted subclass. Review this only as a scoped literature-status packet; do not count it as a new mathematical result or as a solution of that narrower subproblem.

References

- [1] J. Ma and P. Petersen, *Linear independence of compactly supported separable shearlet systems*, arXiv:1404.1690.
- [2] H. Führ and Y. Maus, *Wavelet Riesz Bases Associated to Nonisotropic Dilations*, arXiv:1510.01832.